Function Notation & Evaluations

Full Solutions & Explanations

Digital SAT Style | 50 Questions

QUICK ANSWER KEY

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
Q1	B	Q11	A	Q21	7	Q31	A	Q41	A
Q2	A	Q12	A	Q22	B	Q32	A	Q42	A
Q3	B	Q13	A	Q23	A	Q33	A	Q43	A
Q4	260	Q14	B	Q24	A	Q34	A	Q44	A
Q5	C	Q15	A	Q25	B	Q35	A	Q45	B
Q6	A	Q16	A	Q26	A	Q36	A	Q46	A
Q7	A	Q17	A	Q27	70	Q37	B	Q47	A
Q8	B	Q18	B	Q28	A	Q38	A	Q48	C
Q9	C	Q19	C	Q29	A	Q39	4	Q49	A
Q10	A	Q20	A	Q30	A	Q40	A	Q50	A

Q1. **[EASY]**

Correct Answer: B

STEP-BY-STEP SOLUTION:

Step 1: Substitute x = 5 into f(x) = 7x + 3.

Step 2: f(5) = 7(5) + 3 = 35 + 3 = 38.

WHY EACH OPTION:

A) This is 7 × 5 = 35 without adding the constant 3.

B) Correct. f(5) = 7(5) + 3 = 38.

C) This results from computing 7 × 6 = 42, using the wrong input.

D) This comes from multiplying 5 × 10 + 3, a calculation error.

TRAP ALERT: ■

Forgetting to add the constant term after multiplying.

KEY INSIGHT: ■

f(a) means substitute a everywhere you see x, then simplify.

FAST METHOD: ■

Straight substitution: 7(5) + 3 = 38. No shortcut needed.

Q2. **[EASY]**

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Substitute x = 3: g(3) = 4(3) – 9.

Step 2: g(3) = 12 – 9 = 3.

WHY EACH OPTION:

A) Correct. 4(3) – 9 = 3.

B) This is 4 × 3 = 12, forgetting to subtract 9.

C) This is 4 × 3 + 9 = 21, adding instead of subtracting.

D) This comes from 3 – 9 = –6 then halving, an arithmetic error.

TRAP ALERT: ■

Adding 9 instead of subtracting 9 — sign errors are common.

KEY INSIGHT: ■

Pay close attention to whether the constant is added or subtracted.

FAST METHOD: ■

Direct substitution. 4(3) = 12, then 12 – 9 = 3.

Q3. **[EASY]**

Correct Answer: B

STEP-BY-STEP SOLUTION:

Step 1: Substitute x = –2: h(–2) = –3(–2) + 14.

Step 2: –3(–2) = 6.

Step 3: h(–2) = 6 + 14 = 20.

WHY EACH OPTION:

A) This comes from computing –3(–2) = –6 + 14 = 8, making a sign error on the product.

B) Correct. –3(–2) = 6 and 6 + 14 = 20.

C) This comes from –3(–2) = –6 and –6 – 14 = –20, two sign errors.

D) This comes from 3(2) + 14 = 22, dropping the negative on –3 and using |–2|.

TRAP ALERT: ■

Negative times negative is positive. Students often keep the result negative.

KEY INSIGHT: ■

When substituting a negative value, use parentheses: –3(–2), not –3 · –2 written loosely.

FAST METHOD: ■

–3 × (–2) = +6. Then 6 + 14 = 20.

Q4. [EASY]

Correct Answer: 260

STEP-BY-STEP SOLUTION:

Step 1: Substitute x = 2: f(2) = 150(2) – 40.

Step 2: f(2) = 300 – 40 = 260.

TRAP ALERT: ■

This is a grid-in, so there are no answer choices. Double-check your arithmetic.

KEY INSIGHT: ■

The phrase 'What is the value of f(x) when x = 2' is the same as asking for f(2).

FAST METHOD: ■

150(2) = 300, then 300 – 40 = 260.

Q5. [EASY]

Correct Answer: C

STEP-BY-STEP SOLUTION:

Step 1: Substitute x = 4: f(4) = $(4)^2$ + 5.

Step 2: f(4) = 16 + 5 = 21.

WHY EACH OPTION:

A) This is 4 + 5 = 9, forgetting to square the 4.

B) This is 2(4) + 5 = 13, doubling instead of squaring.

C) Correct. 4^2 + 5 = 16 + 5 = 21.

D) This is $(4 + 5)^2$ – some error, not applying order of operations correctly.

TRAP ALERT: ■

Don't add before squaring. Square x first, then add the constant.

KEY INSIGHT: ■

With quadratics, always square the input first before performing other operations.

FAST METHOD: ■

$4^2 = 16$, plus 5 = 21.

Q6. [EASY]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Substitute x = 7: p(7) = 12 – 2(7).

Step 2: p(7) = 12 – 14 = –2.

WHY EACH OPTION:

A) Correct. 12 – 14 = –2.

B) This comes from |12 – 14| = 2, taking the absolute value.

C) This is –12 – 2(7) = –26, making the 12 negative.

D) This is 12 + 2(7) = 26, adding instead of subtracting.

TRAP ALERT: ■

If the subtraction gives a negative answer, keep it negative. Don't flip the sign.

KEY INSIGHT: ■

When the format is a – bx, the result can be negative when x is large enough.

FAST METHOD: ■

2 × 7 = 14. Then 12 – 14 = –2.

Q7. [EASY]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Substitute x = 20: f(20) = 20/4 + 6.

Step 2: f(20) = 5 + 6 = 11.

WHY EACH OPTION:

A) Correct. 20/4 = 5, then 5 + 6 = 11.

B) This comes from 20/(4 + 6) = 20/10 = 2... error leading to 6.5.

C) This is 20 + 6 = 26, forgetting to divide by 4.

D) This is 20 + 4 + 6 = 30... rounding/computation error.

TRAP ALERT: ■

Divide x by 4 first, then add 6. Don't add 6 to the denominator.

KEY INSIGHT: ■

Follow order of operations: division before addition.

FAST METHOD: ■

20 ÷ 4 = 5, plus 6 = 11.

Q8. [EASY]

Correct Answer: B

STEP-BY-STEP SOLUTION:

Step 1: Substitute $x = 6$: $f(6) = 3(6 + 4)$.

Step 2: $f(6) = 3(10) = 30$.

WHY EACH OPTION:

A) This is $3(6) + 4 = 22$, distributing incorrectly (adding 4 after multiplying only 6).

B) Correct. $6 + 4 = 10$, then $3 \times 10 = 30$.

C) This is $3 \times 6 = 18$, ignoring the + 4 entirely.

D) This is $(3 \times 6) + (3 \times 4)$ + some extra... computation error.

TRAP ALERT: ■

Add inside the parentheses first, then multiply. Or distribute: $3(6) + 3(4) = 18 + 12 = 30$.

KEY INSIGHT: ■

Both approaches work: simplify parentheses first, or distribute. Result is the same.

FAST METHOD: ■

$6 + 4 = 10$. Then $3 \times 10 = 30$.

Q9. [MEDIUM]

Correct Answer: C

STEP-BY-STEP SOLUTION:

Step 1: Substitute $x = –1$: $f(–1) = 2(–1)^2 – 3(–1) + 1$.

Step 2: $(–1)^2 = 1$, so $2(1) = 2$.

Step 3: $–3(–1) = 3$.

Step 4: $f(–1) = 2 + 3 + 1 = 6$.

WHY EACH OPTION:

A) This comes from $2(1) – 3(1) + 1 = 0$, using +1 instead of –1 in the linear term.

B) This comes from $2(–1) – 3(–1) + 1 = –2 + 3 + 1 = 2$... then doubled somehow to 4.

C) Correct. $2(1) + 3 + 1 = 6$.

D) This comes from $2(–1) – 3(1) + 1 = –2 – 3 + 1 = –4$, squaring incorrectly and sign error.

TRAP ALERT: ■

$(–1)^2 = +1$, not –1. And $–3(–1) = +3$, not –3. Two sign traps in one problem.

KEY INSIGHT: ■

When substituting negatives into quadratics, handle the squaring and the linear coefficient signs separately.

FAST METHOD: ■

Checklist: square first → 1, multiply by 2 → 2. Linear: $–3(–1) = +3$. Constant: +1. Sum: 6.

Q10. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Set $g(x) = 17$: $5x – 8 = 17$.

Step 2: Add 8 to both sides: $5x = 25$.

Step 3: Divide by 5: $x = 5$.

WHY EACH OPTION:

A) Correct. $5(5) – 8 = 25 – 8 = 17$. ✓

B) This comes from $17/5 \approx 3.4$, rounding to 3, and ignoring the –8.

C) This is $17 + 8 = 25$, but then forgetting to divide by 5... or $17 – 8 = 9$.

D) This is $17/5 – 8/5 = 9/5 = 1.8$, dividing each term by 5 before combining.

TRAP ALERT: ■

This problem is reversed: you're given the output and must find the input. Set the expression equal to 17 and solve.

KEY INSIGHT: ■

"For what value of x is $g(x) = 17$" means solve the equation $g(x) = 17$ for x.

FAST METHOD: ■

$5x – 8 = 17 \rightarrow 5x = 25 \rightarrow x = 5$.

Q11. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: First find $g(2)$: $g(2) = 2(2) – 1 = 4 – 1 = 3$.

Step 2: Then find $f(3)$: $f(3) = 3(3) + 7 = 9 + 7 = 16$.

WHY EACH OPTION:

A) Correct. $g(2) = 3$, then $f(3) = 16$.

B) This is $f(2) = 3(2) + 7 = 13$, evaluating f at 2 instead of $g(2)$.

C) This is $f(2) + g(2) = 13 + 3 = 16$... no, 19 comes from $f(4) = 19$, skipping the subtraction in g.

D) This is $g(f(2))$ evaluated incorrectly, or $3(2) + 2(2) = 10$.

TRAP ALERT: ■

Work inside-out: evaluate the inner function first, then use that result in the outer function.

KEY INSIGHT: ■

$f(g(2))$ means first compute $g(2)$, then plug that answer into f.

FAST METHOD: ■

Inner first: $g(2) = 3$. Outer: $f(3) = 16$.

Q12. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: f(5) = 25 – 20 + 3 = 8.

Step 2: f(1) = 1 – 4 + 3 = 0.

Step 3: f(5) – f(1) = 8 – 0 = 8.

WHY EACH OPTION:

A) Correct. f(5) = 8 and f(1) = 0, so 8 – 0 = 8.

B) This comes from computing f(5) and f(1) as the same value — arithmetic error.

C) This comes from f(5) = 8 and f(1) = –4 (forgetting the +3), giving 8 – (–4) = 12.

D) This comes from f(5) – f(1) = (5 – 1) = 4, incorrectly thinking the difference of inputs equals the difference of outputs.

TRAP ALERT: ■

You must evaluate each function value separately, then subtract. Do NOT subtract the inputs.

KEY INSIGHT: ■

f(a) – f(b) ≠ f(a – b). Always compute each value independently.

FAST METHOD: ■

f(5) = 25 – 20 + 3 = 8. f(1) = 0. Difference = 8.

Q13. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Substitute x = 3: f(3) = (3 – 3)(3 + 5).

Step 2: f(3) = (0)(8) = 0.

WHY EACH OPTION:

A) Correct. One factor becomes 0, making the whole expression 0.

B) This is $(3 + 5)^2$/... some error, treating (0)(8) as 8 squared.

C) This comes from (3 – 3) = –6... arithmetic error.

D) This is just 3 + 5 = 8, ignoring the first factor.

TRAP ALERT: ■

If any factor equals zero, the entire product is zero. Recognize this immediately.

KEY INSIGHT: ■

x = 3 is a zero (root) of this function because (x – 3) becomes 0.

FAST METHOD: ■

See (x – 3) with x = 3? That factor is 0. Product is 0. Done.

Q14. [MEDIUM]

Correct Answer: B

STEP-BY-STEP SOLUTION:

Step 1: Substitute x = 3: f(3) = |2(3) – 10|.

Step 2: f(3) = |6 – 10| = |–4| = 4.

WHY EACH OPTION:

A) This computes $2(3) - 10 = -4$ but forgets to take the absolute value.

B) Correct. $|-4| = 4$.

C) This is $|2(3) + 10| = 16$, adding instead of subtracting.

D) This is $-|2(3) + 10| = -16$, multiple errors.

TRAP ALERT: ■

Absolute value always returns a non-negative result. If you get a negative, take its positive.

KEY INSIGHT: ■

|expression| means evaluate the expression, then make the result non-negative.

FAST METHOD: ■

$2(3) - 10 = -4$. Absolute value → 4.

Q15. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Replace x with (a + 1): $f(a + 1) = 3(a + 1)^2 + 2$.

Step 2: $(a + 1)^2 = a^2 + 2a + 1$.

Step 3: $3(a^2 + 2a + 1) + 2 = 3a^2 + 6a + 3 + 2 = 3a^2 + 6a + 5$.

WHY EACH OPTION:

A) Correct. $3(a + 1)^2 + 2 = 3a^2 + 6a + 5$.

B) This comes from squaring only the a: $3a^2 + 1 + 2 = 3a^2 + 3$.

C) This is the result before adding the final +2 constant: $3a^2 + 6a + 3$ (forgot the original +2).

D) This comes from $(a + 1)^2 = a^2 + 1$, missing the middle term 2a.

TRAP ALERT: ■

$(a + 1)^2 \neq a^2 + 1$. You must use the full expansion: $a^2 + 2a + 1$.

KEY INSIGHT: ■

When substituting an expression for x, put it in parentheses and expand carefully.

FAST METHOD: ■

Expand $(a+1)^2 = a^2+2a+1$, multiply by 3, add 2.

Q16. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Replace x with 2a: $f(2a) = 6(2a) - 4$.

Step 2: $f(2a) = 12a - 4$.

WHY EACH OPTION:

A) Correct. $6 \times 2a = 12a$, then subtract 4.

B) This ignores the 2 in 2a, using $f(a) = 6a - 4$.

C) This doubles the constant too: 6(2a) – 2(4) = 12a – 8.

D) This factors out 2 from the entire expression, but 2(6a – 4) = 12a – 8, not 12a – 4.

TRAP ALERT: ■

Only the x gets replaced by 2a. The constant –4 stays the same.

KEY INSIGHT: ■

Substitute 2a for x, multiply the coefficient, but leave constants unchanged.

FAST METHOD: ■

6(2a) = 12a. Keep –4. Answer: 12a – 4.

Q17. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Substitute t = 2: h(2) = $-16(2)^2$ + 80(2) + 4.

Step 2: –16(4) = –64.

Step 3: 80(2) = 160.

Step 4: h(2) = –64 + 160 + 4 = 100.

WHY EACH OPTION:

A) Correct. –64 + 160 + 4 = 100.

B) This is –64 + 160 = 96, forgetting to add the constant +4.

C) This is 16(4) + 80(2) + 4 = 64 + 160 + 4 = 228... no, 164 comes from dropping a sign.

D) This is 64 + 160 + 4 = 228, making the –16 positive.

TRAP ALERT: ■

The leading coefficient is negative. –16(4) = –64, not +64.

KEY INSIGHT: ■

In projectile motion problems, the $-16t^2$ term is always negative (it represents gravity pulling down).

FAST METHOD: ■

–16(4) = –64. 80(2) = 160. Total: –64 + 160 + 4 = 100.

Q18. [MEDIUM]

Correct Answer: B

STEP-BY-STEP SOLUTION:

Step 1: Find the slope: m = (20 – 8)/(4 – 0) = 12/4 = 3.

Step 2: The y-intercept is f(0) = 8, so f(x) = 3x + 8.

Step 3: f(6) = 3(6) + 8 = 18 + 8 = 26.

WHY EACH OPTION:

A) This is 4 × 6 = 24, using an incorrect slope of 4.

B) Correct. Slope = 3, y-intercept = 8, so f(6) = 26.

C) This is 20 + 8 = 28, adding the two given outputs.

D) This is 20 + 12 = 32, adding the change (12) to f(4).

TRAP ALERT: ■

Find the slope first from the two points, then build the equation, then evaluate.

KEY INSIGHT: ■

f(0) directly gives you the y-intercept. Use it with the slope to write $f(x) = mx + b$.

FAST METHOD: ■

Slope = $(20–8)/(4–0) = 3$. $f(x) = 3x + 8$. $f(6) = 26$.

Q19. [MEDIUM]

Correct Answer: C

STEP-BY-STEP SOLUTION:

Step 1: $f(x + 3) = 4(x + 3) + 1 = 4x + 12 + 1 = 4x + 13$.

Step 2: $f(x) = 4x + 1$.

Step 3: $f(x + 3) – f(x) = (4x + 13) – (4x + 1) = 12$.

WHY EACH OPTION:

A) This is just the 3 from the input change, not the output change.

B) This comes from 4(3) + 1 – 4(0) – 1 = 12... no, 7 might come from 4 + 3.

C) Correct. The 4x terms cancel, leaving 13 – 1 = 12.

D) This is just the slope (coefficient of x), but 4 × 3 = 12 is the actual change.

TRAP ALERT: ■

For a linear function with slope m, $f(x + h) – f(x) = mh$. Here $m = 4$, $h = 3$, so the change is 12.

KEY INSIGHT: ■

The x-terms always cancel in $f(x+h) – f(x)$ for linear functions, leaving just $m × h$.

FAST METHOD: ■

Slope × change = 4 × 3 = 12. No need to expand fully.

Q20. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Factor f(x): $x^2 – 9 = (x – 3)(x + 3)$.

Step 2: $f(x)/g(x) = (x – 3)(x + 3)/(x + 3)$.

Step 3: Cancel (x + 3): $= x – 3$ (for $x \neq –3$).

WHY EACH OPTION:

A) Correct. Difference of squares factors, then cancel the common factor.

B) This keeps (x + 3) instead of cancelling it.

C) This subtracts 9 from x^2 and divides by x, getting $x – 9/x$, which isn't this.

D) This comes from $x^2/x – 9 = x – 9$, dividing term-by-term incorrectly.

TRAP ALERT: ■

Recognize $x^2 - 9$ as a difference of squares: $(x-3)(x+3)$.

KEY INSIGHT: ■

f(x)/g(x) questions on the DSAT almost always involve factoring and cancelling.

FAST METHOD: ■

$x^2 - 9 = (x-3)(x+3)$. Cancel $(x+3)$. Answer: $x - 3$.

Q21. [MEDIUM]

Correct Answer: 7

STEP-BY-STEP SOLUTION:

Step 1: Set $f(k) = 19$: $2k + 5 = 19$.

Step 2: Subtract 5: $2k = 14$.

Step 3: Divide by 2: $k = 7$.

TRAP ALERT: ■

This is a reverse evaluation: given the output, find the input. Solve the equation.

KEY INSIGHT: ■

$f(k) = 19$ simply means 'plug in k and the result is 19.' Write the equation and solve.

FAST METHOD: ■

$2k + 5 = 19 \rightarrow 2k = 14 \rightarrow k = 7$.

Q22. [MEDIUM]

Correct Answer: B

STEP-BY-STEP SOLUTION:

Step 1: The vertex x-coordinate is $x = -b/(2a) = -6/(2(-1)) = 3$.

Step 2: $g(3) = -(3)^2 + 6(3) = -9 + 18 = 9$.

Step 3: Since $a = -1 < 0$, the parabola opens downward, so 9 is the maximum.

WHY EACH OPTION:

A) This is the coefficient of x, not the maximum value.

B) Correct. Vertex at $x = 3$ gives $g(3) = 9$.

C) This is the x-coordinate of the vertex, not the y-value.

D) This comes from $6 \times 2 = 12$ or $2 \times 6 = 12$, a miscalculation.

TRAP ALERT: ■

Don't confuse the x-coordinate of the vertex (where the max occurs) with the actual maximum value (y-value).

KEY INSIGHT: ■

For $f(x) = ax^2 + bx + c$ with $a < 0$, maximum value $= f(-b/2a)$.

FAST METHOD: ■

$x = -6/(-2) = 3$. $g(3) = -9 + 18 = 9$.

Q23. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: First find f(1): f(1) = 3(1) – 2 = 1.

Step 2: Then find g(1): g(1) = $(1)^2$ = 1.

WHY EACH OPTION:

A) Correct. f(1) = 1, then g(1) = 1.

B) This is f(1) = 3(1) = 3, forgetting to subtract 2, then not squaring.

C) This is f(1) = 3 – 2 = 1, then g(1) = 1... 7 comes from f(3) or some other error.

D) This is g(3) = 9, computing f(1) as 3 (forgetting to subtract 2) then squaring.

TRAP ALERT: ■

Compute the inner function first. f(1) = 1, not 3.

KEY INSIGHT: ■

g(f(1)): start from the innermost parentheses and work outward.

FAST METHOD: ■

f(1) = 1. g(1) = 1. Answer: 1.

Q24. [HARD]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: f(0) = c = 5.

Step 2: f(1) = a + b + 5 = 5, so a + b = 0. ... (i)

Step 3: f(–1) = a – b + 5 = 3, so a – b = –2. ... (ii)

Step 4: Add (i) and (ii): 2a = –2, so a = –1.

WHY EACH OPTION:

A) Correct. Adding the two equations gives 2a = –2, so a = –1.

B) This comes from subtracting the equations instead of adding.

C) This comes from using a + b = 0 and solving a = –b, then miscalculating.

D) This comes from doubling the correct answer due to an arithmetic error.

TRAP ALERT: ■

Use f(0) to find c first. Then use f(1) and f(–1) to create a system for a and b.

KEY INSIGHT: ■

Adding the f(1) and f(–1) equations eliminates b: f(1) + f(–1) = 2a + 2c.

FAST METHOD: ■

f(1) + f(–1) = 2a + 2c → 5 + 3 = 2a + 10 → 2a = –2 → a = –1.

Q25. [HARD]

Correct Answer: B

STEP-BY-STEP SOLUTION:

Step 1: f(f(x)) means substitute f(x) into f.

Step 2: $f(x) = 2x + 3$, so $f(f(x)) = f(2x + 3)$.

Step 3: $f(2x + 3) = 2(2x + 3) + 3 = 4x + 6 + 3 = 4x + 9$.

WHY EACH OPTION:

A) This is $2(2x + 3) = 4x + 6$, forgetting to add the outer +3.

B) Correct. $f(2x + 3) = 2(2x + 3) + 3 = 4x + 9$.

C) This adds 3 twice (3 + 3 = 6? No, 2x + 3 + 3 = 2x + 6...) confusion.

D) This squares the expression $(2x + 3)^2$, which is not what composition does.

TRAP ALERT: ■

f(f(x)) is NOT f(x) × f(x). It's composition: plug f(x) back into f.

KEY INSIGHT: ■

$f(f(x)) = f(2x + 3)$. Replace every x in f with (2x + 3).

FAST METHOD: ■

Outer f does 2(input) + 3. Input = 2x + 3. So $2(2x+3) + 3 = 4x + 9$.

Q26. [HARD]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: $f(g(x)) = 3\cdot g(x) - 1 = 6x + 5$.

Step 2: So $3\cdot g(x) = 6x + 6$.

Step 3: $g(x) = (6x + 6)/3 = 2x + 2$.

WHY EACH OPTION:

A) Correct. $3(2x + 2) - 1 = 6x + 6 - 1 = 6x + 5$. ✓

B) This gives $f(g(x)) = 3(2x+1) - 1 = 6x + 2$, not $6x + 5$.

C) This gives $f(g(x)) = 3(2x+5) - 1 = 6x + 14$, not $6x + 5$.

D) This gives $f(g(x)) = 3(3x+2) - 1 = 9x + 5$, wrong coefficient of x.

TRAP ALERT: ■

Don't guess — set up the equation $f(g(x)) = 3\cdot g(x) - 1$ and solve for g(x).

KEY INSIGHT: ■

If f(g(x)) is given and f is known, replace g(x) with a variable, solve, and that variable is g(x).

FAST METHOD: ■

$3\cdot g(x) - 1 = 6x + 5 \rightarrow 3\cdot g(x) = 6x + 6 \rightarrow g(x) = 2x + 2$.

Q27. [EASY]

Correct Answer: 70

STEP-BY-STEP SOLUTION:

Step 1: Substitute x = 1: $f(1) = 85(1) - 15$.

Step 2: $f(1) = 85 - 15 = 70$.

TRAP ALERT: ■

Simple substitution. Watch the subtraction.

KEY INSIGHT: ■

When $x = 1$, $f(1)$ is just the sum of all coefficients and constants.

FAST METHOD: ■

$85 – 15 = 70$.

Q28. [EASY]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: $f(0) = –5(0) + 20 = 0 + 20 = 20$.

WHY EACH OPTION:

A) Correct. When $x = 0$, only the constant term remains.

B) This assumes $f(0) = 0$, confusing 'input is 0' with 'output is 0.'

C) This is the slope, not $f(0)$.

D) This comes from $–5 + 20$... but $–5(0) = 0$, not $–5$.

TRAP ALERT: ■

$f(0)$ is the y-intercept. It equals the constant term when the function is linear.

KEY INSIGHT: ■

For any function $f(x) = mx + b$, $f(0) = b$.

FAST METHOD: ■

$f(0)$ = constant term = 20.

Q29. [EASY]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: $g(15) = 15/3 – 4 = 5 – 4 = 1$.

WHY EACH OPTION:

A) Correct. $15 \div 3 = 5$, then $5 – 4 = 1$.

B) This is $15/3 = 5$, forgetting to subtract 4.

C) This is $15 – 4 = 11$, forgetting to divide by 3.

D) This might come from a rounding or sign error.

TRAP ALERT: ■

Divide first, then subtract. Follow the order of operations.

KEY INSIGHT: ■

Division before subtraction. $g(15) = 15/3 – 4$, not $15/(3–4)$.

FAST METHOD: ■

$15/3 = 5$. Then $5 – 4 = 1$.

Q30. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: $g(5) = 5 - 4 = 1$.

Step 2: $f(1) = 1^2 + 2(1) = 1 + 2 = 3$.

WHY EACH OPTION:

A) Correct. $g(5) = 1$, then $f(1) = 3$.

B) This is $f(5) = 25 + 10 = 35$, evaluating f at 5 instead of $g(5)$.

C) This comes from $f(g(5)) = f(5) - 4 = 35 - 4 = 31$... wrong, or some other error leading to 8.

D) This comes from $g(4) = 0$, then $f(0) = 0$, using the wrong input.

TRAP ALERT: ■

Always compute the inner function first: $g(5) = 1$, then plug 1 into f.

KEY INSIGHT: ■

$f(g(5))$: evaluate g at 5 first, then f at that result.

FAST METHOD: ■

$g(5) = 1$. $f(1) = 1 + 2 = 3$.

Q31. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Set $f(x) = 0$: $3x^2 - 12 = 0$.

Step 2: $3x^2 = 12$.

Step 3: $x^2 = 4$.

Step 4: $x = \pm 2$. The positive value is $x = 2$.

WHY EACH OPTION:

A) Correct. $x^2 = 4 \rightarrow x = 2$ (positive root).

B) This is $x^2 = 4$, but then $x = 4$ (confusing $x^2 = 4$ with $x = 4$).

C) This comes from $3x = 12 \rightarrow x = 4$... wait, 6 comes from $12/2 = 6$, different error.

D) This is just the constant 12, not a solution.

TRAP ALERT: ■

When $x^2 = 4$, $x = 2$, not $x = 4$. Don't confuse $x^2 = k$ with $x = k$.

KEY INSIGHT: ■

Zeros of f mean $f(x) = 0$. Solve the resulting equation for x.

FAST METHOD: ■

$3x^2 = 12 \rightarrow x^2 = 4 \rightarrow x = 2$ (positive).

Q32. [HARD]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: f(5) = (2(5) – 1)/(5 + 3) = (10 – 1)/8 = 9/8.

WHY EACH OPTION:

A) Correct. Numerator: 10 – 1 = 9. Denominator: 5 + 3 = 8. Result: 9/8.

B) This is (2(5) + 1)/8 = 11/8, adding 1 instead of subtracting.

C) This flips the fraction: 8/9 instead of 9/8.

D) This comes from simplifying incorrectly or using wrong values.

TRAP ALERT: ■

Evaluate numerator and denominator separately, then divide. Don't simplify prematurely.

KEY INSIGHT: ■

For rational functions, compute top and bottom independently, then form the fraction.

FAST METHOD: ■

Top: 2(5)–1 = 9. Bottom: 5+3 = 8. Answer: 9/8.

Q33. [HARD]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: f(2) = $(2)^3$ – $2(2)^2$ + 2 – 3.

Step 2: = 8 – 2(4) + 2 – 3.

Step 3: = 8 – 8 + 2 – 3 = –1.

WHY EACH OPTION:

A) Correct. 8 – 8 + 2 – 3 = –1.

B) This comes from 8 – 8 + 2 – 3 = –1, then taking |–1| = 1.

C) This is just the constant term –3, ignoring all other terms.

D) This comes from 8 – 8 + 2 + 3 = 5... no, 3 likely from a sign error.

TRAP ALERT: ■

Carefully compute each term. 2^3 = 8 and 2^2 = 4, not 2^3 = 6.

KEY INSIGHT: ■

For polynomials with multiple terms, evaluate each power of x separately, then combine.

FAST METHOD: ■

8 – 8 + 2 – 3 = –1. The first two terms cancel, leaving –1.

Q34. [EASY]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: f(10) = 10 – 10 = 0.

WHY EACH OPTION:

A) Correct. 10 – 10 = 0.

B) This substitutes incorrectly or doesn't subtract.

C) This is 10 + 10 = 20, adding instead of subtracting.

D) This is –10, possibly from 0 – 10.

TRAP ALERT: ■

Straightforward substitution. Make sure you subtract x from 10, not 10 from x.

KEY INSIGHT: ■

f(x) = 10 – x means 'start with 10, subtract whatever x is.'

FAST METHOD: ■

10 – 10 = 0.

Q35. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: f(x) = 4x – 7.

Step 2: 2·f(x) = 2(4x – 7) = 8x – 14.

Step 3: 2·f(x) + 3 = 8x – 14 + 3 = 8x – 11.

WHY EACH OPTION:

A) Correct. 2(4x – 7) + 3 = 8x – 14 + 3 = 8x – 11.

B) This is 2·f(x) = 8x – 14, but forgets to add the +3.

C) This comes from 2(4x) – 7 + 3 = 8x – 4, not distributing the 2 to –7.

D) This ignores the –7 entirely: 2(4x) + 3 = 8x + 3.

TRAP ALERT: ■

Distribute the 2 to BOTH terms of f(x), then add 3.

KEY INSIGHT: ■

Scaling a function: multiply the entire expression, including the constant.

FAST METHOD: ■

2(4x – 7) + 3 = 8x – 14 + 3 = 8x – 11.

Q36. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: f(a) = 5a + 2 and f(b) = 5b + 2.

Step 2: 5a + 2 = (5b + 2) + 10.

Step 3: 5a + 2 = 5b + 12.

Step 4: 5a = 5b + 10.

Step 5: a = b + 2.

WHY EACH OPTION:

A) Correct. Dividing $5a = 5b + 10$ by 5 gives $a = b + 2$.

B) This comes from $a = b + 10$, not dividing 10 by the coefficient 5.

C) This comes from $a = b + 10/2 = b + 5$, dividing by 2 instead of 5.

D) This substitutes incorrectly, not using function notation properly.

TRAP ALERT: ■

Don't forget to divide the +10 by the slope (5) when isolating a.

KEY INSIGHT: ■

For linear functions, a difference of 10 in outputs corresponds to a difference of 10/m in inputs.

FAST METHOD: ■

Output diff = 10, slope = 5, so input diff = 10/5 = 2. Thus $a = b + 2$.

Q37. [HARD]

Correct Answer: B

STEP-BY-STEP SOLUTION:

Step 1: $f(x) = x^2 - 6x + 8$.

Step 2: $f(x + 4) = (x+4)^2 - 6(x+4) + 8 = x^2 + 8x + 16 - 6x - 24 + 8 = x^2 + 2x$.

Step 3: Set equal: $x^2 - 6x + 8 = x^2 + 2x$.

Step 4: $-6x + 8 = 2x \rightarrow 8 = 8x \rightarrow x = 1$.

Step 5: Exactly one value of x.

WHY EACH OPTION:

A) Incorrect. There is exactly one solution: $x = 1$.

B) Correct. The x^2 terms cancel, leaving a linear equation with one solution.

C) This assumes a quadratic equation remains, but the x^2 terms cancel.

D) This would only be true if the equation became $0 = 0$, an identity.

TRAP ALERT: ■

When you expand and set equal, the x^2 terms cancel, leaving a LINEAR equation.

KEY INSIGHT: ■

$f(x) = f(x+4)$ is about symmetry of the parabola. The axis of symmetry determines the solution.

FAST METHOD: ■

The axis of symmetry is $x = 3$. $f(x) = f(x+4)$ when x and x+4 are symmetric about $x = 3$: $x = 1$.

Q38. [EASY]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: $f(3) = 200(3) + 50 = 600 + 50 = 650$.

Step 2: Since f(x) gives total cost for x hours, $f(3) = 650$ means 3 hours costs $650.

WHY EACH OPTION:

A) Correct. $x = 3$ is the input (hours), $f(3) = 650$ is the output (cost).

B) This reverses input and output.

C) This interprets the result as a rate, not a total cost.

D) This confuses units and reverses the roles.

TRAP ALERT: ■

Match input to what x represents and output to what f(x) represents.

KEY INSIGHT: ■

Interpretation questions require understanding: input → x, output → f(x), and what each means in context.

FAST METHOD: ■

x = hours, f(x) = cost. f(3) = 650 → 3 hours costs $650.

Q39. [MEDIUM]

Correct Answer: 4

STEP-BY-STEP SOLUTION:

Step 1: Set equal: $x^2 – 8 = 2x$.

Step 2: $x^2 – 2x – 8 = 0$.

Step 3: $(x – 4)(x + 2) = 0$.

Step 4: $x = 4$ or $x = –2$. Positive value: $x = 4$.

TRAP ALERT: ■

Set the two functions equal and solve the resulting equation. Factor if possible.

KEY INSIGHT: ■

Where two functions are equal = where their graphs intersect.

FAST METHOD: ■

$x^2 – 2x – 8 = 0 \to (x – 4)(x + 2) = 0 \to x = 4$ (positive).

Q40. [HARD]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Expand: $(x + 2)^2 – 9 = x^2 + 4x + 4 – 9 = x^2 + 4x – 5$.

Step 2: Factor: $x^2 + 4x – 5 = (x + 5)(x – 1)$.

Step 3: The zeros are $x = –5$ and $x = 1$, which appear as constants in $(x + 5)(x – 1)$.

WHY EACH OPTION:

A) Correct. $(x+5)(x–1) = x^2 + 4x – 5$. ✓

B) This gives $x^2 – 4x – 5$, with the wrong middle term.

C) This is $x^2 – 9$, not $(x+2)^2 – 9$.

D) This is the expanded form, but doesn't display the zeros as constants — it's standard form, not factored.

TRAP ALERT: ■

The question asks for the form that shows the zeros. Factored form shows zeros; standard form doesn't.

KEY INSIGHT: ■

'Displays zeros as constants' = factored form. The zeros are where each factor = 0.

FAST METHOD: ■

Expand to x^2+4x-5, factor to $(x+5)(x-1)$.

Q41. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Set $f(x) = x$: $3x - 5 = x$.

Step 2: $2x = 5$.

Step 3: $x = 5/2$.

WHY EACH OPTION:

A) Correct. $3(5/2) - 5 = 15/2 - 5 = 5/2$. ✓

B) This comes from $3x = 5 + x \rightarrow 2x = 5 \rightarrow x = 5/2$... wait, 5 comes from not subtracting x.

C) This comes from $4x = 5$, perhaps adding x instead of subtracting.

D) This is negative, from a sign error.

TRAP ALERT: ■

$f(x) = x$ means the function's output equals its input. Set the expression equal to x and solve.

KEY INSIGHT: ■

Fixed points of a function occur where $f(x) = x$.

FAST METHOD: ■

$3x - 5 = x \rightarrow 2x = 5 \rightarrow x = 5/2$.

Q42. [HARD]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: Test $f(0) = 3$: $(0)^2 + 3 = 3$. ✓

Step 2: Test $f(1) = 4$: $(1)^2 + 3 = 4$. ✓

Step 3: Test $f(2) = 7$: $(2)^2 + 3 = 7$. ✓

Step 4: Test $f(-1) = 4$: $(-1)^2 + 3 = 4$. ✓

Step 5: Test $f(-2) = 7$: $(-2)^2 + 3 = 7$. ✓

Step 6: All values match. Answer: $f(x) = x^2 + 3$.

WHY EACH OPTION:

A) Correct. All five values match.

B) $f(2) = 2(4) + 3 = 11 \neq 7$.

C) $f(0) = -3 \neq 3$.

D) $f(1) = 3 + 3 = 6 \neq 4$.

TRAP ALERT: ■

Plug in at least 2–3 values to verify. Starting with f(0) quickly eliminates wrong options.

KEY INSIGHT: ■

Table-matching questions: test $x = 0$ first (easiest), then test another value to confirm.

FAST METHOD: ■

f(0) eliminates C (gives –3). f(1): A gives 4 ✓, B gives 5 ✗, D gives 6 ✗. Answer: A.

Q43. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: $g(x) = f(x - 3) + 2$.

Step 2: Since $f(x) = x^2$, replace x with $(x - 3)$: $f(x - 3) = (x - 3)^2$.

Step 3: So $g(x) = (x - 3)^2 + 2$.

WHY EACH OPTION:

A) Correct. Direct substitution of (x–3) into f.

B) This expands and simplifies incorrectly.

C) This uses (x + 3) instead of (x – 3), a common sign error.

D) This ignores the transformation and just adds constants: $3 + 2 = 5$.

TRAP ALERT: ■

$f(x - 3)$ means shift right by 3, not left. The sign inside is opposite the direction.

KEY INSIGHT: ■

$f(x - h) + k$ shifts the graph right by h and up by k.

FAST METHOD: ■

Replace x in f with (x – 3), then add 2.

Q44. [HARD]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: $f(2) = 2a + b = 11$. ... (i)

Step 2: $f(5) = 5a + b = 23$. ... (ii)

Step 3: Subtract (i) from (ii): $3a = 12$, so $a = 4$.

Step 4: From (i): $2(4) + b = 11 \rightarrow b = 3$.

Step 5: $f(0) = a(0) + b = 3$.

WHY EACH OPTION:

A) Correct. $a = 4$, $b = 3$, $f(0) = 3$.

B) This comes from $23 - 11 = 12$, then 12/... some miscalculation giving $b = 5$.

C) This comes from subtracting a from b: $4 - 3 = 1$.

D) This comes from adding: $4 + 3 = 7$.

TRAP ALERT: ■

$f(0) = b$ (the y-intercept). Find a first from the two equations, then find b.

KEY INSIGHT: ■

Two points determine a linear function. Use the slope formula, then solve for the intercept.

FAST METHOD: ■

Slope = (23–11)/(5–2) = 4. Then b = 11 – 2(4) = 3. f(0) = 3.

Q45. [EASY]

Correct Answer: B

STEP-BY-STEP SOLUTION:

Step 1: f(x) = 8 is a constant function. It equals 8 for all values of x.

Step 2: f(100) = 8.

WHY EACH OPTION:

A) This confuses the input with the output.

B) Correct. A constant function always outputs 8, regardless of input.

C) This multiplies 8 × 100, but the function doesn't multiply.

D) This adds 8 + 100, but the function doesn't add.

TRAP ALERT: ■

f(x) = 8 means the output is always 8. The value of x doesn't matter.

KEY INSIGHT: ■

A constant function has no x in its rule. f(anything) = 8.

FAST METHOD: ■

Constant function → output = 8. Done.

Q46. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: f(3) = 2(3) = 6.

Step 2: f(5) = 2(5) = 10.

Step 3: f(3) + f(5) = 6 + 10 = 16.

Step 4: Check: f(8) = 2(8) = 16. ✓

WHY EACH OPTION:

A) Correct. f(3) + f(5) = 16 = f(8).

B) f(6) = 12 ≠ 16.

C) f(15) = 30 ≠ 16.

D) f(7) = 14 ≠ 16.

TRAP ALERT: ■

f(a) + f(b) ≠ f(a + b) in general, but for f(x) = 2x (linear through origin), it does.

KEY INSIGHT: ■

Only for functions of the form f(x) = cx (passing through the origin) does f(a) + f(b) = f(a+b).

FAST METHOD: ■

Compute each: $f(3) = 6$, $f(5) = 10$, sum = 16. $f(8) = 16$. ✓

Q47. [HARD]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: $f(4) = (4 - 1)^2 + k = 15$.

Step 2: $(3)^2 + k = 15$.

Step 3: $9 + k = 15$.

Step 4: $k = 6$.

WHY EACH OPTION:

A) Correct. $9 + k = 15 \rightarrow k = 6$.

B) This is the value of $(4-1)^2 = 9$, not k.

C) This is $4 - 1 = 3$, confusing the base with the answer.

D) This is $15 + 9 = 24$, adding instead of subtracting.

TRAP ALERT: ■

Substitute $x = 4$, set equal to 15, then solve for k. Don't confuse $(4-1)^2$ with k.

KEY INSIGHT: ■

When a function contains an unknown constant, use a given input-output pair to find it.

FAST METHOD: ■

$(4-1)^2 = 9$. $9 + k = 15$. $k = 6$.

Q48. [HARD]

Correct Answer: C

STEP-BY-STEP SOLUTION:

Step 1: $f(4) = 2(3)^4 = 2(81) = 162$.

Step 2: $f(3) = 2(3)^3 = 2(27) = 54$.

Step 3: $f(4) - f(3) = 162 - 54 = 108$.

WHY EACH OPTION:

A) This is $f(4) - f(3) = 2$... thinking the function is linear.

B) This is $f(3) = 54$, not the difference.

C) Correct. $162 - 54 = 108$.

D) This is $f(4) = 162$, not the difference.

TRAP ALERT: ■

For exponential functions, the difference between consecutive outputs is NOT constant.

KEY INSIGHT: ■

$f(4) - f(3)$ for exponential functions grows — it's not the same as $f(1) - f(0)$.

FAST METHOD: ■

2·81 = 162. 2·27 = 54. Difference = 108.

Q49. [MEDIUM]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: $f(x) = -2x + 10$.

Step 2: $f(-x) = -2(-x) + 10 = 2x + 10$.

Step 3: Set equal: $-2x + 10 = 2x + 10$.

Step 4: $-4x = 0$.

Step 5: $x = 0$.

WHY EACH OPTION:

A) Correct. The 10s cancel, leaving $-2x = 2x$, so $x = 0$.

B) This is the y-intercept ÷ 2, or where $f(x) = 0$.

C) This is the negative of the x-intercept.

D) This is the y-intercept itself.

TRAP ALERT: ■

$f(-x)$ means replace x with $-x$ everywhere. Then set $f(x) = f(-x)$ and solve.

KEY INSIGHT: ■

$f(x) = f(-x)$ defines an even function. For a linear function (other than constant), the only solution is $x = 0$.

FAST METHOD: ■

$-2x + 10 = 2x + 10 \rightarrow -4x = 0 \rightarrow x = 0$.

Q50. [HARD]

Correct Answer: A

STEP-BY-STEP SOLUTION:

Step 1: First find $g(2)$: $g(2) = 2(2) - 3 = 1$.

Step 2: Then $h(2) = f(g(2)) = f(1) = (1)^2 + 1 = 2$.

WHY EACH OPTION:

A) Correct. $g(2) = 1$, $f(1) = 2$.

B) This is $f(2) = 4 + 1 = 5$, evaluating f at 2 directly.

C) This comes from $f(g(2))$ with $g(2) = 3$ (adding instead of subtracting): $f(3) = 10$.

D) This is $f(2) - 1 = 4$ or $g(2) + 3 = 4$.

TRAP ALERT: ■

Compute g first, then f. Don't evaluate f at the original input.

KEY INSIGHT: ■

$h(x) = f(g(x))$ is composition. Always work inside-out.

FAST METHOD: ■

$g(2) = 1$. $f(1) = 2$. Done.

TOPIC SUMMARY — Function Notation & Evaluations

Common Mistakes:

1. Sign errors when substituting negative values — especially $(–x)^2 = +x^2$, not $–x^2$.
2. Confusing f(a + b) with f(a) + f(b) — these are NOT the same for most functions.
3. In composition f(g(x)), evaluating f first instead of the inner function g.

Must Remember:

1. f(a) means substitute a for x in the expression defining f, then simplify.
2. f(g(x)) means evaluate g first, then use that result as the input for f (inside-out).
3. "For what value of x is f(x) = k" means solve the equation f(x) = k for x.

Expected on DSAT:

Function notation and evaluation questions typically appear 2–4 times per test across both modules. They range from simple direct substitution (Module 1) to composition, reverse evaluation, and algebraic manipulation involving function notation (Module 2, harder section). This is one of the most frequently tested topics in the Algebra domain.